

List of scientific publications by Breno Cordeiro Bispo

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Required information: **Publication(s) resulting from the doctoral dissertation: A1, A4, A5, B1, C1, D2, D3.**

* = corresponding author; # = shared first authorship

A) Peer-Reviewed: published or accepted for publishing

1. **Bispo, B.C.***, de Oliveira Neto, J.R., Lima, J.B., Santos, F.A.N. (2026). From pairwise to higher-order brain community detection: A hypergraph signal processing approach on brain functional connectivity analysis. *Computers in Biology and Medicine*, 201, 111409. <https://doi.org/10.1016/j.combiomed.2025.111409>
2. Lima, J.B., **Bispo, B.C.**, Silva Neto, E.F., Morgado, P. (2026). Novel measures of pairwise time series interdependence using graph permutation entropies: An application to brain functional connectivity. *Computers in Biology and Medicine*, 209, 111682. <https://doi.org/10.1016/j.combiomed.2026.111682>
3. Sardellitti, S., **Bispo, B.C.**, Santos, F.A.N., Lima, J.B. (2025). Cross-Laplacians Based Topological Signal Processing over Cell Multicomplexes. *Proceedings of the 33rd European Signal Processing Conference (EUSIPCO)*, Palermo, Italy, pp. 2647–2651. <https://doi.org/10.23919/EUSIPCO63237.2025.11226604>
4. **Bispo, B.C.***, Sardellitti, S., Santos, F.A.N., Lima, J.B. (2025). Learning Higher-Order Interactions in Brain Networks via Topological Signal Processing. *Proceedings of the 33rd European Signal Processing Conference (EUSIPCO)*, Palermo, Italy, pp. 2652–2656. <https://doi.org/10.23919/EUSIPCO63237.2025.11226759>
5. **Bispo, B.C.***, Santos, F.A.N., de Oliveira Neto, J.R., Lima, J.B. (2024). Emergence of Higher-Order Functional Brain Connectivity with Hypergraph Signal Processing. *Proceedings of the 32nd European Signal Processing Conference (EUSIPCO)*, Lyon, France, pp. 1332–1336. <https://doi.org/10.23919/EUSIPCO63174.2024.10715376>
6. **Bispo, B.C.**, Cavalcante, E.L., Rodrigues, M.A.B. (2024): Access Control System Integrated with RFID and NFC-Enabled Smartphone Technologies. In: *IFMBE Proceedings*, 9th ed., Eds. Springer Nature Switzerland, v. 100, p. 657–667. https://doi.org/10.1007/978-3-031-49407-9_65
7. **Bispo, B.C.***, Cunha, N.A., Silva, M.C.B.C., Rodrigues, M.A.B. (2024): Batteryless NFC Device for Heart Rate and SpO2 Acquisition. In: *IFMBE Proceedings*, 9th ed., Eds. Springer Nature Switzerland, v. 100, p. 465–475. https://doi.org/10.1007/978-3-031-49407-9_47
8. Silva, A.T.D., **Bispo, B.C.**, Esteves, G.R.P., Silva, M.C.B.C., Rodrigues, M.A.B. (2022): A Serious Game that Can Aid Physiotherapy Treatment in Children Using Dennis Brown Orthotics. In: *XXVII Brazilian Congress on Biomedical Engineering – IFMBE Proceedings*, Eds. Springer (Cham), v. 83, p. 445–450. https://doi.org/10.1007/978-3-030-70601-2_69
9. **Bispo, B.C.***, Cavalcante, E.L., Esteves, G.R.P., Silva, M.C.B.C., Alves, G.J., Rodrigues, M.A.B. (2022): Access Control in Hospitals with RFID and BLE Technologies. In: *XXVII Brazilian Congress on Biomedical Engineering – IFMBE Pro-*

- ceedings, Eds. Springer (Cham), v.83, p.917–924. https://doi.org/10.1007/978-3-030-70601-2_137
10. Silva, A.T.D., **Bispo, B.C.**, Conceição, A.M., Silva, M.C.B.C., Esteves, G.R.P., Rodrigues, M.A.B. (2022): Adapted Children’s Serious Game Using Dennis Brown Orthotics During the Preparatory Phase for the Gait. In: *XXVII Brazilian Congress on Biomedical Engineering – IFMBE Proceedings*, Eds. Springer (Cham), v.83, p.439–443. https://doi.org/10.1007/978-3-030-70601-2_68
 11. Cunha, N.A., **Bispo, B.C.**, Cavalcante, E.L., Inocencio, A.V.M., Alves, G.J., Rodrigues, M.A.B. (2022): Application of MQTT Network for Communication in Healthcare Establishment. In: *XXVII Brazilian Congress on Biomedical Engineering – IFMBE Proceedings*, Eds. Springer (Cham), v.83, p.1465–1470. https://doi.org/10.1007/978-3-030-70601-2_216
 12. Silva, M.C.B.C., **Bispo, B.C.**, Silva, C.M., Cunha, N.A., Santos, E.A.B., Rodrigues, M.A.B. (2022): Assisted Navigation System for the Visually Impaired. In: *XXVII Brazilian Congress on Biomedical Engineering – IFMBE Proceedings*, Eds. Springer (Cham), v.83, p.1439–1444. https://doi.org/10.1007/978-3-030-70601-2_212
 13. Cunha, N.A., **Bispo, B.C.**, Ferreira, K.R.C., Alves, G.J., Esteves, G.R.P., Santos, E.A.B., Rodrigues, M.A.B. (2022): Communication in Hospital Environment Using Power Line Communications. In: *XXVII Brazilian Congress on Biomedical Engineering – IFMBE Proceedings*, Eds. Springer (Cham), v.83, p.1451–1455. https://doi.org/10.1007/978-3-030-70601-2_214
 14. Silva, A.T.D., Silveira, J.V.B.D., Esteves, G.R.P., **Bispo, B.C.**, Cunha, N.A., Kunkel, M.E., Rodrigues, M.A.B. (2022): Development and Customization of a Dennis Brown Orthosis Prototype Produced from Anthropometric Measurements by Additive Manufacturing. In: *XXVII Brazilian Congress on Biomedical Engineering – IFMBE Proceedings*, Eds. Springer (Cham), v.83, p.465–470. https://doi.org/10.1007/978-3-030-70601-2_72
 15. Chaves Filho, A.C., Inocencio, A.V.M., Ferreira, K.R.C., Cavalcante, E.L., **Bispo, B.C.**, Rodrigues, C.M.B., Lessa, P.S., Rodrigues, M.A.B. (2022): Geriatric Physiotherapy: A Telerehabilitation System for Identifying Therapeutic Exercises. In: *XXVII Brazilian Congress on Biomedical Engineering – IFMBE Proceedings*, Eds. Springer (Cham), v.83, p.969–974. https://doi.org/10.1007/978-3-030-70601-2_144
 16. Silva, C.M., **Bispo, B.C.**, Esteves, G.R.P., Cavalcante, E.L., Oliveira, A.L.B., Silva, M.C.B.C., Cunha, N.A., Rodrigues, M.A.B. (2022): Transducer for the Strengthening of the Pelvic Floor Through Electromyographic Biofeedback. In: *XXVII Brazilian Congress on Biomedical Engineering – IFMBE Proceedings*, Eds. Springer (Cham), v.83, p.935–940. https://doi.org/10.1007/978-3-030-70601-2_139
 17. **Bispo, B.C.***, Cunha, N.A., Cavalcante, E.L., Esteves, G.R.P., Ferreira, K.R.C., Chaves Filho, A.C., Rodrigues, M.A.B. (2022): Wearable System for Postural Adaptation. In: *XXVII Brazilian Congress on Biomedical Engineering – IFMBE Proceedings*, Eds. Springer (Cham), v.83, p.387–393. https://doi.org/10.1007/978-3-030-70601-2_60

B) Monographs and Edited Volumes

1. **Bispo, B.C.** (2026). *Higher-Order Signal Processing and Its Applications in Brain Network Analysis*. Federal University of Pernambuco, Recife. (Doctoral dissertation, Graduate Program in Electrical Engineering)
2. **Bispo, B.C.** (2021): *Vital Sign Monitoring via NFC*. Federal University of Pernambuco, Recife. (Master's thesis, Graduate Program in Electrical Engineering)
3. **Bispo, B.C.** (2019): *Access Control System via RFID/NFC*. Federal University of Pernambuco, Recife, (Undergraduate final project, Undergraduate Program in Electronic engineering)

C) Submitted Manuscripts with Peer-Review

1. **Bispo, B.C.***, Sardellitti, S., Santos, F.A.N., Lima, J.B. (2026). Multimodal Higher-Order Brain Networks: A Topological Signal Processing Perspective. Submitted to: *IEEE Journal of Biomedical and Health Informatics* (publisher's acknowledgement of receipt enclosed). Earlier version, see <https://arxiv.org/abs/2603.29903>.
2. Lima, J.B., Silva Neto, E.F., **Bispo, B.C.**, Morgado, P. (2026). Schizophrenia Detection from Electroencephalogram Signals Using Graph Permutation Entropies. Submitted to: *Biomedical Signal Processing and Control* (publisher's acknowledgement of receipt enclosed).
3. Silva Neto, E.F., **Bispo, B.C.**, Lima, J.B. (2026). Tensor-Based Hypergraph Learning for Schizophrenia and ADHD Classification. Submitted to: *IEEE Transactions on Image Processing* (publisher's acknowledgement of receipt enclosed).
4. Sardellitti, S.#, **Bispo, B.C.#**, Santos, F.A.N., Lima, J.B. (2025). Topological Signal Processing Over Cell MultiComplexes Via Cross-Laplacian Operators. Submitted to: *IEEE Transactions on Signal Processing* (publisher's acknowledgement of receipt enclosed). Earlier version, see <https://arxiv.org/abs/2510.09139>.

D) Further published contributions to the Scientific Community

1. **Bispo, B.C.** (2026). Network Inference. Talk presented at: *CIMPA Research School – The Mathematics of Complex Systems: Theory and Applications*. Recife, Brazil.
2. **Bispo, B.C.**, Sardellitti, S., Santos, F.A.N., Lima, J.B. (2025). Application of Topological Signal Processing to Brain Networks. Talk presented at: [Cutting-Edge Perspectives on Brain Networks](#). Universitas Mercatorum, Rome, Italy.
3. **Bispo, B.C.***, Santos, F.A.N., de Oliveira Neto, J.R., Lima, J.B. (2024): Emergence of Higher-Order Functional Brain Connectivity with Hypergraph Signal Processing. In: *Graph Signal Processing Workshop 2024* (extended abstract). Delft, Netherlands.

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